

SYSTEM ARCHITECTURE DOCUMENT

LKA | AEB | Endurance

Team Equinox

aBAJA SAEINDIA 2026 — Software Virtual World Simulation

1. System Overview

This document describes the unified software architecture of Team Equinox's autonomous vehicle platform for aBAJA SAEINDIA 2026, covering all three dynamic events: Lane Keep Assist (LKA), Autonomous Emergency Braking (AEB), and Endurance. The system operates within IPG CarMaker 15.0, communicates via the CarMaker Python APO interface, and uses only camera and radar RSI sensors — no ground-truth lane data or DVA lane quantities are used as algorithm inputs.

Parameter	Value	Notes
Simulator	IPG CarMaker 15.0	Primary simulation and vehicle dynamics platform
Rendering	MovieNX	Camera RSI visual rendering engine
Camera Sensor	Camera RSI — 640×480 RGB	~30 Hz over TCP port 2210; primary LKA sensor
Radar Sensor	Automotive Radar RSI	Distance + velocity output; primary AEB sensor
Target Speed	30 km/h	Set in config.py, enforced via CarMaker APO
Control Loop Rate	30 Hz	Unified rate for both LKA and AEB pipelines
Vehicle Model	Demo_IPG_CompanyCar	Wheelbase 2.622 m, mass 1,391 kg
Language	Python 3	OpenCV, NumPy, IPG CarMaker Python API

2. Module Architecture

The stack is decomposed into eight Python modules with strict downward dependencies. The main loop imports all sub-modules; no sub-module imports the main loop. LKA and AEB pipelines share the same control cycle but operate on independent sensor streams.

Module	Event Scope	Role	Key Outputs
Equinox_Main.py	LKA + AEB + Endurance	Master control loop — initialises APO, runs 30 Hz cycle, arbitrates between LKA steering and AEB braking commands, manages endurance state machine.	Steering cmd (rad), Brake cmd (%)
perception_engine.py	LKA	BEV warp, Sliding Window lane search, polynomial fitting, lateral error and lookahead computation.	Lateral error (m), lookahead (m)
radar_processor.py	AEB	Processes radar RSI output: range and relative velocity estimation, TTC calculation, obstacle classification.	Range (m), TTC (s), threat flag
aeb_controller.py	AEB	Fuzzy logic braking controller: maps TTC and closing speed to braking intensity. Issues brake command and AEB event log.	Brake intensity (%), AEB flag
equinox_config.py	All	Single source of truth for all tunable parameters: PID gains, BEV coords,	Constants (imported)

		fuzzy logic membership functions, endurance thresholds.	
equinox_utils.py	All	RMSE accumulator, DNF watchdog, endurance lap/time tracker, unified CSV telemetry logger.	RMSE, DNF flag, lka_run.csv, aeb_log.csv
tcp_client.py	LKA	TCP socket client for Camera RSI stream on localhost:2210. Frame decoding and auto-reconnect.	Raw BGR frame (NumPy array)
radar_client.py	AEB	TCP/UDP socket client for Radar RSI data stream. Parses range, velocity, and bearing packets.	Radar data struct

Event 1 — Lane Keep Assist (LKA)

3.1 LKA Data Flow

Each LKA control cycle (nominally 1/30 s) executes the following 13-step pipeline:

#	Module	Description
1	tcp_client	Receive raw 640×480 RGB frame from Camera RSI over TCP port 2210
2	perception_engine	Convert to grayscale → apply binary thresholding
3	perception_engine	Warp to Bird's-Eye View (source trapezoid → top-down rectangle)
4	perception_engine	Bottom-half histogram → identify left/right lane base pixel positions
5	perception_engine	Sliding window search upward (9 windows, ±50 px margin) → collect lane pixels
6	perception_engine	Fit 1st-order polynomial to each lane → compute lane centre
7	perception_engine	Compute lateral error = lane centre – image centre (converted to metric)
8	perception_engine	Compute lookahead offset at 5.0 m ahead on fitted lane curve
9	Equinox_Main	Pure Pursuit: compute baseline steering angle from lookahead geometry
10	Equinox_Main	PID: compute correction term from lateral error ($K_p=0.10$, $K_i=0.005$, $K_d=0.08$)
11	Equinox_Main	Sum Pure Pursuit + PID → apply rate-limit (0.03 rad/frame) → clamp to ±0.20 rad
12	Equinox_Main	Write road-wheel angle to CarMaker APO (DVA write)
13	equinox_utils	Log frame telemetry to CSV; update RMSE accumulator; check DNF watchdog

3.2 Perception Pipeline — BEV + Sliding Window

3.2.1 Bird's-Eye View Transform

A perspective warp maps a trapezoidal region of the front-facing camera image to a top-down rectangle. This eliminates perspective foreshortening so lane lines appear as near-vertical strips regardless of distance. A pixels-per-metre calibration ratio converts pixel offsets to metric lateral errors.

3.2.2 Sliding Window Lane Search

A bottom-half column histogram identifies lane origins. A stack of 9 windows (± 50 px margin) searches upward, re-centring on the mean of detected white pixels at each level. This naturally bridges dashed lane segments and handles curved roads without angle assumptions.

3.2.3 Polynomial Fitting and Gap Bridging

A 1st-order polynomial is fitted to the detected pixel set in each window stack. Lane centre and lookahead point are derived from the mean of both fitted curves. Polynomial fits are cached for up to 15 frames, allowing stable steering through brief detection gaps caused by dashed lines or partial occlusion.

3.3 LKA Control Architecture — Pure Pursuit + PID

A standalone PID controller reacts to lateral error but has no knowledge of road curvature, making it prone to oscillation. Pure Pursuit provides a geometrically grounded baseline — it computes the exact steering angle to follow an arc to the lookahead point using the vehicle wheelbase and steering ratio. The PID term compensates residual steady-state error.

Parameter	Value	Rationale
Lookahead Distance	5.0 m	~0.6 s preview at 30 km/h; balances responsiveness and smoothness
Vehicle Wheelbase	2.622 m	Demo_IPG_CompanyCar specification — used directly in Pure Pursuit formula
Steering Ratio	16:1	Maps road-wheel angle to virtual steering wheel angle
Max Road-Wheel Angle	± 0.20 rad	Physical steering limit enforced in software to prevent spin-out
Rate Limit	0.03 rad/frame	Prevents actuator saturation and jerky motion; maintains tire grip
PID Kp / Ki / Kd	0.10 / 0.005 / 0.08	Empirically tuned; Ki eliminates steady-state offset; Kd damps overshoot

Event 2 — Autonomous Emergency Braking (AEB)

4.1 AEB System Overview

The AEB subsystem detects obstacles using an automotive radar sensor and computes braking commands through a fuzzy logic controller. It operates in parallel with the LKA pipeline within the same 30 Hz control loop, with Equinox_Main arbitrating between steering and braking outputs so both systems can respond simultaneously without conflict.

4.2 AEB Data Flow

#	Module	Description
1	radar_client	Receive radar RSI packet: raw range, relative velocity, and bearing data
2	radar_processor	Parse and validate packet → extract obstacle range (m) and closing speed (m/s)
3	radar_processor	Compute Time-to-Collision (TTC) = range ÷ closing speed
4	radar_processor	Classify obstacle threat level: CLEAR / CAUTION / CRITICAL based on TTC thresholds
5	aeb_controller	Map TTC and closing speed to fuzzy membership functions (near/mid/far, slow/medium/fast)
6	aeb_controller	Apply fuzzy inference rules → defuzzify to braking intensity (0–100%)
7	Equinox_Main	If braking intensity > 0%: override throttle; write brake command to CarMaker APO
8	equinox_utils	Log AEB event to aeb_log.csv: timestamp, range, TTC, braking intensity, threat class

4.3 Radar Processing

The radar_processor module converts raw radar RSI output into actionable threat data. Range and relative velocity are extracted from each packet. TTC is computed assuming constant relative velocity. Obstacle classification uses three threat zones:

Zone	TTC Range	Closing Speed	System Response
CLEAR	> 3.0 s	Any	No braking. Normal LKA operation continues uninterrupted.
CAUTION	1.5 – 3.0 s	> 2.0 m/s	Partial braking applied. Fuzzy logic outputs 20–60% braking intensity.
CRITICAL	< 1.5 s	Any	Full emergency braking. Fuzzy logic outputs 60–100%. AEB flag raised.

4.4 Fuzzy Logic Braking Controller

The aeb_controller module implements a Mamdani-type fuzzy inference system. The two inputs (TTC and closing speed) are fuzzified, processed through a rule base, and defuzzified using centroid method to produce a crisp braking intensity percentage.

Fuzzy Input	Membership Functions	Description
Time-to-Collision (TTC)	FAR (>3 s) / MID (1.5–3 s) / NEAR (<1.5 s)	Classifies temporal proximity to obstacle
Closing Speed	SLOW (<2 m/s) / MEDIUM (2–5 m/s) / FAST (>5 m/s)	Classifies rate of approach
Braking Intensity (out)	NONE / PARTIAL / FULL (0% / 20–60% / 60–100%)	Defuzzified braking command

The fuzzy controller is preferred over a threshold-only system because it produces smooth, graduated braking commands that avoid sudden actuator jumps — reducing the risk of rear-end collision from abrupt deceleration.

4.5 AEB Safety Parameters

Parameter	Value	Rationale
TTC CRITICAL threshold	< 1.5 s	Minimum reaction window at 30 km/h with system latency factored in
TTC CAUTION threshold	< 3.0 s	Provides graduated warning zone for partial braking response
Max braking intensity	100%	Full brake command sent to CarMaker APO in CRITICAL zone
AEB loop rate	30 Hz	Matched to camera loop; both pipelines share the same control tick
Radar reconnect	Auto	radar_client reconnects automatically on socket drop

Event 3 — Endurance

5.1 Endurance Overview

The Endurance event requires the vehicle to complete a sustained autonomous run combining both LKA and AEB over an extended course. Both subsystems run concurrently and continuously within the same 30 Hz control loop. The equinox_utils module manages an endurance state machine, continuous telemetry logging, RMSE tracking, and the DNF watchdog throughout the run.

5.2 Unified Control Loop — LKA + AEB Concurrent Operation

Equinox_Main runs a single unified control cycle at 30 Hz. Each tick executes both the LKA perception pipeline and the AEB radar pipeline, then applies a priority arbitration scheme before issuing actuator commands:

Priority	Condition	Action
1 — Emergency Brake	AEB: CRITICAL zone (TTC < 1.5 s)	Override LKA steering. Apply full braking. Suspend steering output for duration of event.
2 — Caution Brake	AEB: CAUTION zone (TTC 1.5–3.0 s)	Continue LKA steering. Apply partial braking (20–60%) simultaneously.
3 — LKA Only	AEB: CLEAR zone (TTC > 3.0 s)	Normal LKA operation. No braking applied. Steering command issued as computed.

The priority scheme ensures that a critical obstacle event always takes precedence over lane correction. After the CRITICAL condition clears, the LKA pipeline resumes automatically without manual reset.

5.3 Endurance State Machine

The endurance state machine in equinox_utils tracks the overall health and progress of the run across five states:

State	Trigger	System Behaviour
INITIALISING	System start	APO connection established; sensor streams verified; all modules loaded.
RUNNING	All checks pass	Unified LKA+AEB loop active at 30 Hz. Telemetry logging active.
AEB_ACTIVE	AEB CRITICAL or CAUTION flag raised	Braking command issued. LKA suspended (CRITICAL) or continued (CAUTION). Event logged.
LANE_LOST	Lane detection fails > 15 frames	Fit cache fallback active. DNF timer increments. Warning logged.
DNF	Front axle outside lane > 5 s	DNF event logged. Steering set to 0. Run halted per competition rules.

5.4 Telemetry and Logging

All run data is written to results/EQUINOX_ALL/ in real time. The logging subsystem runs within equinox_utils and does not introduce latency into the 30 Hz control loop.

File / Folder	Description
Data/Logs/lka_run.csv	Per-frame LKA telemetry: timestamp, lateral error, steering angle, speed, yaw rate, RMSE
Data/Logs/aeb_log.csv	Per-AEB-event log: timestamp, range, TTC, closing speed, braking intensity, threat class
Data/Logs/endurance.csv	Endurance session log: state transitions, DNF timer value, lap time, system uptime
debug_frames/	Per-frame BEV and sliding window images — archived post-run for offline analysis
debug_raw_frame.png	Last raw camera frame received from CarMaker
debug_bev.png	Last Bird's-Eye View frame after perspective warp
debug_vision.png	Last annotated front-view frame with lane centre, lateral error, and sliding window overlay

6. Unified Safety Mechanisms

Seven independent safety mechanisms protect both the vehicle and the integrity of the run across all three dynamic events. All mechanisms are active simultaneously whenever the system is in RUNNING or AEB_ACTIVE state.

Mechanism	Implementation	Event Scope	Behaviour
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DNF Watchdog	equinox_utils.py	LKA / Endurance	If front axle exits lane for >5 consecutive seconds, logs DNF and halts run per competition rules.
Steering Rate Limiter	Equinox_Main.py	LKA / Endurance	Caps steering delta at 0.03 rad/frame to prevent actuator saturation and maintain tire grip.
Angle Saturation Clamp	Equinox_Main.py	LKA / Endurance	Final road-wheel command clamped to ± 0.20 rad — matches physical steering limit of vehicle model.
Fit Cache Fallback	perception_engine.py	LKA / Endurance	Uses last valid polynomial fit for up to 15 frames if lane detection is lost; prevents zero-steering events.
AEB Priority Arbitrator	Equinox_Main.py	AEB / Endurance	CRITICAL braking overrides LKA steering. Restores normal LKA operation automatically when threat clears.
TCP Reconnect — Camera	tcp_client.py	LKA / Endurance	Auto-reconnects to Camera RSI stream on socket drop; exponential backoff prevents reconnect storms.
TCP Reconnect — Radar	radar_client.py	AEB / Endurance	Auto-reconnects to Radar RSI stream on socket drop; same backoff logic as camera client.